Probabilistic Models

Study Notes – Working Draft

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Based on the lecture *Probabilistic Models* (G. Widmer, JKU Linz), WS 2026/27

These notes try to make probabilistic models *click*. Every idea is introduced in plain words first, then as a formula, and whenever things get abstract there is a small example with actual numbers. All worked numbers were re-computed independently. Use the notes as a study aid alongside the lecture slides, not as a replacement for them.

Working draft: Chapters 1–3 are written out. Chapters 4–8 are still outlines and will be filled in as the lecture progresses.

If you spot an error or a gap, please open an issue or pull request at **github.com/steveTgit/AI_Master** — corrections are always welcome.

Good study materials should be shared, not gatekept. If this helped you, take it as a reason to help the next colleague you see struggling with something.

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1

Introduction: Why Probabilistic Models?

Almost everything interesting that an intelligent system does involves information that is incomplete, noisy, or simply wrong. A sensor flickers, a symptom has five possible causes, a microphone picks up the audience coughing. This course is about one systematic answer to the question *how should a machine reason when it cannot be sure?* The answer: describe the world with probabilities, organise those probabilities with a graph, and let general-purpose algorithms do the reasoning.

This first chapter sets the scene. We look at the kind of problems we want to solve, why logic alone does not get us there, and which three big problems the rest of the course is going to tackle.

1.1 Reasoning Under Uncertainty

The lecture opens with a handful of scenarios that look very different on the surface:

- **Diagnosing a complex system.** A spacecraft (or one of its subsystems) sends sensor signals that indicate errors — "Houston, we have a problem". Which fault most likely caused them? What would happen if we tried a certain repair? Which repair should we actually do?
- **Autonomous driving.** Given a road map and streams of GPS, camera, radar and range-sensor data, drive a car through difficult terrain, with other cars and obstacles around.
- **Aircraft tracking.** From intermittent, sometimes occluded radar or video signals: where will each aircraft be in 0.1, 1 or 10 seconds, how many aircraft are there, and which one is which?
- **Speech recognition.** Split an audio stream into words, transcribe them, and use spoken commands to control devices.
- **Real-time music following.** Listen to a live performance and keep track of where the musicians currently are in the printed score — for example to turn pages automatically or to synchronise other events with the music.

Underneath, all of them have the same structure. An autonomous system receives information about its environment, at regular or irregular intervals. From that information it has to identify objects, classify situations, make predictions and take decisions. And the information is incomplete, uncertain, noisy, sometimes wrong and occasionally contradictory.

That last point is what breaks purely logical or deterministic approaches. Try writing down the rule "muscle pain ⇒ flu". It is wrong for most people with muscle pain. Patch it with exceptions ("... unless they went to the gym, unless they slept badly, unless ...") and the list never ends.

What we actually know is softer: *muscle pain makes flu more likely*, and we would like to say *how much* more likely. That is exactly what probabilities are for.

Probabilistic graphical models give us four things at once. They

- represent knowledge about the world *and* about its uncertainty,
- support inference: drawing conclusions and making decisions,
- can be learned from example observations, and
- technically, are graphical representations of *factorised* probability distributions.

The last point will only make sense after Chapter 3; by then it should feel obvious. The field sits where artificial intelligence, statistics, probability theory, machine learning and algorithms meet.

1.2 The Declarative Approach

There are two ways to build a system that solves problems. The *procedural* way is to write a special-purpose algorithm for each problem. The *declarative* way — the classic AI philosophy, and the one this course follows — is to

1. build a *model* of the problem or world you want to reason about, and
2. write *general inference algorithms* that work on any such model and draw conclusions from it.

This buys you a lot. Knowledge (the model) and reasoning (the algorithm) are cleanly separated. The model has a clear, well-defined meaning, independent of whatever algorithm happens to run on it. The inference algorithms are generic: they do not care which domain the model describes. And if the world changes, you only change the model, not the code.

Intuition & Mental Model

Think of a city map versus a set of written directions. Directions (procedural) get you along *one* route; if a street is closed, they are useless. A map (declarative) answers *any* routing question with one general skill — reading a map — and if a street closes, you update the map, not every set of directions. A probabilistic model is the map; inference algorithms are the map-reading skill.

1.3 What Are Probabilistic Graphical Models?

A probabilistic graphical model is a formalism for describing the structure of a domain and for quantifying how uncertain we are about it. It is called *probabilistic* because it represents a probability distribution, and *graphical* because the dependencies between the variables are made explicit in a graph.

Three questions run through the whole course:

Representation (modelling). How do we represent the relevant aspects of a world and the connections between them?

Inference. Given incomplete, uncertain, noisy observations and the model, how do we reach conclusions about things we cannot observe, or about the future?

Learning. How do we build models of complex worlds automatically from (lots of) observations?

Table 1.1 shows how the chapters of these notes map to the lecture units. The course covers *directed* models only: static Bayesian networks and temporal models (dynamic Bayesian networks, hidden Markov models and the Kalman filter). Undirected and semi-directed models (Markov networks, chain graphs) are not part of the course.

Chapter	Topic	Lecture units
1	Introduction	VO 1
2	Fundamentals of probability: inference and independence	VO 1–2
3	Bayesian networks: representation and semantics	VO 3
4	Inference in Bayesian networks: exact, approximate, MCMC	VO 4–5
5	Learning Bayesian networks: parameters and structure	VO 6–8
6	Temporal models: dynamic Bayesian networks	VO 9
7	Hidden Markov models: inference, learning, applications	VO 10–11
8	Continuous linear systems: the Kalman filter	video

Table 1.1: Roadmap of the course. Representation, inference and learning first for static models (Chapters 3–5), then for models that evolve over time (Chapters 6–8).

1.4 A First Example: A Tiny Diagnosis World

To make this concrete, consider a toy medical diagnosis setting. There are two possible diseases, *flu* and *hayfever*; two possible symptoms, (sinus) *congestion* and *muscle pain*; and the *season*, which might also matter. We want a system that answers questions like:

- How likely is it that a patient has the flu, given that it is fall and she has a sinus congestion but no muscle pain?
- How likely is it that a patient does *not* have hayfever, given that she has muscle pain and the flu?
- In which season are people most likely to have a congestion but no hayfever?

The probabilistic approach takes three steps. First, model every relevant entity as a *random variable*. Second, model the relationships and influences between them as (conditional) probabilities. Third, treat question answering as *computing conditional probabilities*, given whatever has been observed. Here, five variables are enough:

$$\begin{aligned} \mathit{Flu},\ \mathit{Hayfever},\ \mathit{Congestion},\ \mathit{MusclePain} &\in \{\mathit{true}, \mathit{false}\}, \\ \mathit{Season} &\in \{\mathit{spring}, \mathit{summer}, \mathit{fall}, \mathit{winter}\}, \end{aligned}$$

so the model is defined over $\mathcal{X} = \{\mathit{Flu}, \mathit{Hayfever}, \mathit{Congestion}, \mathit{MusclePain}, \mathit{Season}\}$.

Implicitly, we assume that the variables describe the state of the world (one specific patient) at one point in time, that this state is completely characterised by the values of the variables, that every variable always takes exactly one of its values, and that some of these values may be known to us while others are hidden.

The second part of the model is the *full joint distribution*: one probability for every possible combination of values. It is a table that sums to one and here has $2 \times 2 \times 2 \times 2 \times 4 = 64$ entries. The first few rows might look like this:

$$\begin{aligned} P(\neg \mathit{flu}, \neg \mathit{hayfever}, \neg \mathit{congestion}, \neg \mathit{musclepain}, \mathit{spring}) &= 0.014 \\ P(\mathit{flu}, \neg \mathit{hayfever}, \neg \mathit{congestion}, \neg \mathit{musclepain}, \mathit{spring}) &= 0.003 \\ P(\neg \mathit{flu}, \mathit{hayfever}, \neg \mathit{congestion}, \neg \mathit{musclepain}, \mathit{spring}) &= 0.008 \\ &\vdots \end{aligned}$$

The good news, shown in Chapter 2: *every* question of the type above is a conditional probability query, such as $P(Flu = true \mid Season = spring, MusclePain = false)$, and every such query can be answered from the full joint distribution with one universal algorithm. The bad news comes in three flavours:

- **Representational complexity.** The joint table grows exponentially with the number of variables. With 50 binary variables it has about 10^{15} entries — far too many to store.
- **Computational complexity.** Answering a query exactly from the table also takes exponential time.
- **Modelling complexity.** Even if we could store the table, who would give us all those numbers? Building a valid model of a complex world by hand is hard.

The rest of the course is about fighting these three problems. Against the first, we study *(conditional) independence*, use it to break the joint distribution into a product of small pieces ("factorise" it), and draw the result as a compact graph that shows the (in)dependencies — and often the causal structure — of the domain (Fig. 1.1). Against the second, we design algorithms that compute answers directly from the graph, optimise them, and develop fast approximate algorithms. Against the third, we *learn* models from observations, which is where machine learning comes in. All of this needs a solid grip on basic probability theory, which is what Chapter 2 is for.

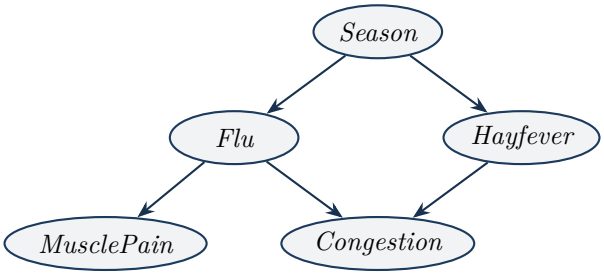


Figure 1.1: A graphical model for the diagnosis world. The season influences both diseases; flu causes muscle pain and congestion; hayfever causes congestion. With this structure the joint distribution needs only 17 numbers instead of 63 (we will count them in Section 3.5).

2

Fundamentals of Probability

Everything later in the course is built from a surprisingly small toolbox: random variables, joint distributions, conditioning, a couple of rules for rewriting probabilities, and — most importantly — *independence*. This chapter fixes the notation used in the lecture, refreshes the basics, and then makes two big points. First, *any* probabilistic question can be answered from the full joint distribution with one simple algorithm, inference by enumeration. Second, that algorithm is hopelessly expensive unless we exploit (conditional) independencies — which is the whole motivation for graphical models.

If you are comfortable with probability, skim the first sections, but read Sections 2.5 and 2.7 carefully: they are the foundation for everything that follows.

2.1 Random Variables and Events

Random variable

A *random variable* X is a variable with a fixed (finite or infinite) domain $Val(X)$, the set of values it can take. It represents one aspect of the world (or of the system's internal state).

Random variables are the atomic building blocks of all our models. They come in three flavours:

- **Boolean** variables take values in $\{false, true\}$ (or $\{0, 1\}$).
- **Discrete** (categorical) variables have a finite set of symbolic values, e.g. $Val(Season) = \{spring, summer, fall, winter\}$.
- **Continuous** variables take real values. Most real-world sensors (temperature, sonar, radar, ...) produce these; we get to them in Section 2.8.

The running example of this chapter is the diagnosis world from Section 1.4: two diseases (not directly observable), two symptoms (possibly observable) and the season (observable), with exactly one patient at a time. Notice the modelling choice: each disease and each symptom gets its *own* Boolean variable. A single variable $Disease \in \{flu, hayfever, none\}$ would be unable to express a patient who has *both* diseases.

Events and atomic events

An *event* is an assignment of values to some or all of the variables of the world. An *atomic event* is an event in which *every* variable has a value assigned.

An atomic event is one complete, fully specified state of the world. A general event fixes only some variables and therefore stands for the *set* of all world states that agree on those values. In the diagnosis world:

- $Season = spring \wedge MusclePain = true$ covers all situations in which it is spring and the patient has muscle pain, whatever her diseases or congestion;
- $Congestion = false$ is an event too;
- $Season = fall \wedge MusclePain = true \wedge Congestion = false \wedge Hayfever = false \wedge Flu = false$ is an atomic event.

Altogether there are $2 \times 2 \times 2 \times 2 \times 4 = 64$ atomic events. Events are what the system computes probabilities for.

Intuition & Mental Model

Picture the 64 atomic events as the 64 cells of one big spreadsheet, each cell holding a probability. Every event is simply a *selection of cells*, and its probability is the sum over the selected cells. Marginals, conditionals and queries later in this chapter are all just clever ways of selecting and summing cells.

2.1.1 Notation

The lecture uses a fixed set of conventions (mostly those of Koller & Friedman); Table 2.1 collects them. It is worth learning them early — the exam uses the same notation.

Example	Meaning
$\mathcal{X}$	the complete set of variables a model is defined over
$A,\ X_i,\ E_j$	a single random variable (e.g. $X_i \in \mathcal{X}$)
$\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathbf{E}, \ldots$	sets of random variables, e.g. $\mathbf{X} = \{X_1, X_2, X_3\}$
$Val(X)$	domain of X: the set of values X can take
$x^0, x^1, \ldots$	the values of X; x^i is the i-th value in $Val(X)$
$x^0,\ x^1$ (Boolean)	by convention $x^0 = false$, $x^1 = true$
$\mathbf{x}, \mathbf{e}, \ldots$	a joint assignment of values to a set, e.g. $\mathbf{E} = \mathbf{e}$
$Val(\mathbf{X})$	all joint assignments to $\mathbf{X}$: $Val(X_1) \times Val(X_2) \times \cdots$
$\mathbf{x}\langle\mathbf{Y}\rangle$	assignment $\mathbf{x}$ restricted to the variables in $\mathbf{Y} \subset \mathbf{X}$
$X_i^{(0)}, X_i^{(1)}, X_i^{(t)}$	the template variable X_i at time t (temporal models, Chapter 6)
$\boldsymbol{\theta}$	the set of parameters of a model
$P(X = x^i)$	a probability (value of the probability mass function)
$P(x^i),\ P(x^i, y^j)$	shorthand for $P(X = x^i)$ and $P(X = x^i, Y = y^j)$
$P(X, Y)$	a probability distribution over the discrete variables X, Y
$p(x)$	a probability *density* function for a continuous X

Table 2.1: Notational conventions of the lecture. For named Boolean variables the slides also write *flu* for $Flu = true$ and $\neg$*flu* for $Flu = false$.

2.2 Probability Distributions

Probability distribution

Let S be a set of mutually exclusive atomic events. A *probability distribution* over S is a function $P : S \to \mathbb{R}$ with

1. $P(\alpha) \geq 0$ for all $\alpha \in S$ (non-negativity)
2. $P(\Omega) = 1$, where $\Omega = \bigcup_{\alpha_i \in S} \alpha_i$ is the disjunction of all possible events (normalisation)
3. if $\alpha, \beta \in S$ and $\alpha \cap \beta = \emptyset$, then $P(\alpha \cup \beta) = P(\alpha) + P(\beta)$ (additivity)

The *probability* $P(\alpha)$ is the value the distribution assigns to the event α. Here $\cap$ and $\cup$ denote logical AND and OR between events.

A tiny example with only two Boolean variables, *Flu* and *MP* (muscle pain), has four atomic events, and one possible distribution over them is:

$P(Flu, MP)$	$MP = false$	$MP = true$
$Flu = false$	0.70	0.15
$Flu = true$	0.05	0.10

So the probability that a randomly chosen person has the flu *and* muscle pain is $P(\mathit{flu}, \mathit{mp}) = 0.1$, and so on.

Probability or distribution? Look at the argument

The same letter P is used for two different kinds of objects:

- if the argument is an **event** (values fixed), it is a *probability* — a single number: $P(\mathit{flu}) = 0.15$;
- if the argument is a (set of) **random variable(s)**, it is a *distribution* — a whole table of numbers: $P(Flu) = (0.85, 0.15)$.

Mixed cases follow the same rule. $P(Flu \mid \mathit{mp})$ is one distribution over *Flu*, whereas $P(Flu \mid MP)$ is a *family* of distributions, one for every value of *MP*.

2.2.1 What does a probability mean?

What exactly "$P(\alpha) = 0.3$" means is a deep philosophical question. Intuitively, $P(\alpha)$ quantifies how confident we are that α is true or will happen: $P(\alpha) = 1$ means we are certain it will, $P(\alpha) = 0$ that it will not. Everything in between needs an interpretation, and there are two main camps:

- **Frequentist:** $P(\alpha)$ is the fraction of times α would occur if we repeated the experiment infinitely often. Fine for dice, awkward for "the probability of rain *this afternoon* is 0.7" — this afternoon happens only once.
- **Subjectivist (Bayesian):** $P(\alpha)$ is a degree of belief that α is true, or the degree to which the available evidence supports α.

The course adopts the subjectivist view: $P(x)$ is the *system's degree of belief* that x holds in the world. When we *learn* probabilities from data (Chapter 5), we will happily fall back on frequencies.

2.3 Joint and Marginal Distributions

Full joint and marginal distributions

In a world described by the variables $\mathcal{X}$:

- the *full joint distribution* over $\mathcal{X}$ is the distribution over all atomic events, i.e. it assigns a probability to every combination of values of all variables;
- a *marginal distribution* over a subset $\mathbf{X} \subset \mathcal{X}$ assigns probabilities to all value combinations of just the variables in $\mathbf{X}$;
- the *marginal distribution of a variable* $X \in \mathcal{X}$ is the special case $\mathbf{X} = \{X\}$.

A few notational shortcuts: we write $P(X = x, Y = y)$ or just $P(x, y)$ for the probability of the conjunction $(X = x) \cap (Y = y)$; $P(\mathbf{X}) = P(X_1, \ldots, X_k)$ for the joint distribution over a set; $P(\mathbf{X}, \mathbf{Y})$ for the joint over $\mathbf{X} \cup \mathbf{Y}$; and $P(\mathbf{X} \mid \mathbf{Y})$ for the distribution over $\mathbf{X}$ conditioned on values of $\mathbf{Y}$ (next section).

The workhorse example of the slides is a diagnosis world with three Boolean variables, $\mathcal{X} = \{Flu, MP, Fever\}$, and this full joint distribution:

	mp		$\neg mp$	
$P(Flu, MP, Fever)$	$fever$	$\neg fever$	$fever$	$\neg fever$
flu	0.15	0.10	0.10	0.03
$\neg flu$	0.03	0.04	0.05	0.50

(2.1)

All eight entries sum to 1, as they must.

Marginals are obtained by *summing out* everything you are not interested in. To get $P(flu)$, add up all cells in the *flu* row — whatever the values of *MP* and *Fever*:

$$P(flu) = \sum_{x,y} P(flu, MP = x, Fever = y) = 0.15 + 0.10 + 0.10 + 0.03 = 0.38.$$

All three marginals of table (2.1)

$$P(\neg flu) = 0.03 + 0.04 + 0.05 + 0.50 = 0.62 \qquad P(Flu) = (0.62,\ 0.38)$$
$$P(\neg mp) = 0.10 + 0.03 + 0.05 + 0.50 = 0.68 \qquad P(MP) = (0.68,\ 0.32)$$
$$P(\neg fever) = 0.10 + 0.04 + 0.03 + 0.50 = 0.67 \qquad P(Fever) = (0.67,\ 0.33)$$

Each distribution is written as $(P(x^0), P(x^1))$, i.e. *false first*.

Why is summing allowed? The event *flu* is the disjunction ($\cup$) of all atomic events with $Flu = true$, and atomic events are mutually exclusive, so by additivity their probabilities simply add up. The same argument works for *any* event: *the probability of an event α is the sum over all atomic events (entries of the full joint) that match α*, i.e. that agree with α on the variables α fixes. This one sentence is already the core of a general inference algorithm (Section 2.5).

2.4 Conditional Probability, the Chain Rule and Bayes' Rule

Most questions we ask a model have the form "I have observed β — what is the probability of the (unobservable) thing α that I care about?" For instance: given that my patient has a congestion, how likely is hayfever, $P(\textit{Hayfever} = \textit{true} \mid \textit{Congestion} = \textit{true})$? And if I also learn that she has muscle pain, how does that change my belief? Or, for a robot: given the current GPS and radar readings, what is the distribution over its possible locations and velocities?

Conditional probability

For two events α, β with $P(\beta) > 0$, the *conditional probability* of α given that β is true is

$$P(\alpha \mid \beta) = \frac{P(\alpha \cap \beta)}{P(\beta)}.$$

The picture behind this formula (Fig. 2.1): learning that β is true rules out every world in which β is false. What remains is the region β, and inside it the worlds where α also holds. We simply rescale so that the surviving region has total probability one.

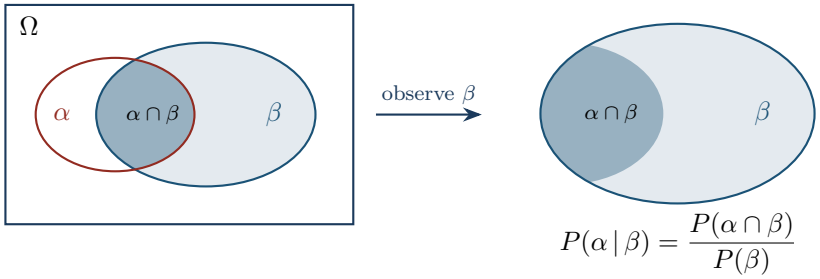


Figure 2.1: Conditioning as zooming in. Observing β throws away all worlds outside β; the region β becomes the new universe and is rescaled to total probability one. $P(\alpha \mid \beta)$ is the share of β that is also α.

Two consequences are worth stating explicitly. For a fixed event β and some variables $\mathbf{X}$, the conditional distribution $P(\mathbf{X} \mid \beta)$ has all properties of a proper distribution — in particular it sums to one over all value combinations of $\mathbf{X}$. And conditioning is an *operation*: it takes a distribution $P(\mathbf{X})$ and returns a new one, $P(\mathbf{X} \mid \beta)$.

In the two-variable table above, $P(\textit{flu}) = 0.05 + 0.10 = 0.15$, but

$$P(\textit{flu} \mid \textit{mp}) = \frac{P(\textit{flu}, \textit{mp})}{P(\textit{mp})} = \frac{0.10}{0.15 + 0.10} = 0.4.$$

Learning about muscle pain raises the belief in flu from 15 % to 40 %.

2.4.1 The chain rule

Rearranging the definition of conditional probability gives the *product rule*, $P(\alpha \cap \beta) = P(\alpha \mid \beta)\, P(\beta)$. Because $\cap$ is symmetric, we may just as well write $P(\alpha \cap \beta) = P(\alpha)\, P(\beta \mid \alpha)$. Applying the product rule over and over to n events gives the chain rule.

Chain rule

$$\begin{aligned} P(\alpha_1, \ldots, \alpha_n) &= P(\alpha_1 \mid \alpha_2, \ldots, \alpha_n)\, P(\alpha_2, \ldots, \alpha_n) \\ &= P(\alpha_1 \mid \alpha_2, \ldots, \alpha_n)\, P(\alpha_2 \mid \alpha_3, \ldots, \alpha_n) \cdots P(\alpha_{n-1} \mid \alpha_n)\, P(\alpha_n). \end{aligned}$$

Since conjunction is commutative and associative, any order works, e.g.

$$P(\alpha_1, \ldots, \alpha_n) = P(\alpha_1)\, P(\alpha_2 \mid \alpha_1) \cdots P(\alpha_n \mid \alpha_1, \ldots, \alpha_{n-1}).$$

It also works partially,

$$P(\alpha_1, \ldots, \alpha_n) = P(\alpha_1, \ldots, \alpha_k)\, P(\alpha_{k+1}, \ldots, \alpha_n \mid \alpha_1, \ldots, \alpha_k),$$

and for whole distributions:

$$P(X_1, \ldots, X_N) = P(X_1, \ldots, X_k)\, P(X_{k+1}, \ldots, X_N \mid X_1, \ldots, X_k).$$

The chain rule is *always* true — it makes no assumptions whatsoever. It just says you can tell the story of a joint event one piece at a time, each new piece conditioned on everything told so far. On its own it does not save anything: the last factor still conditions on all the other variables. The savings come later, when independence lets us shorten those conditioning sets (Section 2.7 and Chapter 3).

2.4.2 The law of total probability

Law of total probability

For any event $\mathbf{X} = \mathbf{x}$ and any set of variables $\mathbf{Y}$:

$$P(\mathbf{x}) = \sum_{\mathbf{y} \in Val(\mathbf{Y})} P(\mathbf{x}, \mathbf{y}) = \sum_{\mathbf{y} \in Val(\mathbf{Y})} P(\mathbf{x} \mid \mathbf{y})\, P(\mathbf{y}).$$

The same holds for whole distributions and with extra conditioning:

$$P(\mathbf{X}) = \sum_{\mathbf{y}} P(\mathbf{X} \mid \mathbf{y})\, P(\mathbf{y}), \qquad P(\mathbf{X} \mid \mathbf{Z}) = \sum_{\mathbf{y}} P(\mathbf{X}, \mathbf{y} \mid \mathbf{Z}) = \sum_{\mathbf{y}} P(\mathbf{X} \mid \mathbf{y}, \mathbf{Z})\, P(\mathbf{y} \mid \mathbf{Z}).$$

The first form is exactly the "summing out" we used for marginals — the law of total probability is what justifies marginalisation and, below, inference by enumeration.[1] The second form is a *case analysis*: split the world into the cases $\mathbf{Y} = \mathbf{y}$, compute the probability of $\mathbf{x}$ in each case, and average, weighting each case by how likely it is. With the two-variable table: $P(mp) = P(mp \mid flu)\, P(flu) + P(mp \mid \neg flu)\, P(\neg flu) = \frac{0.10}{0.15} \cdot 0.15 + \frac{0.15}{0.85} \cdot 0.85 = 0.25$.

2.4.3 Bayes' rule

Write the product rule both ways round, $P(\alpha \cap \beta) = P(\alpha \mid \beta)\, P(\beta) = P(\beta \mid \alpha)\, P(\alpha)$, and divide by $P(\beta)$:

[1] The slides also list $P(\mathbf{x}) = \sum_{\mathbf{y}} P(\mathbf{x})\, P(\mathbf{y} \mid \mathbf{x})$. That is true but trivial, because $\sum_{\mathbf{y}} P(\mathbf{y} \mid \mathbf{x}) = 1$.

Bayes' rule

$$P(\alpha \mid \beta) = \frac{P(\beta \mid \alpha)\, P(\alpha)}{P(\beta)}.$$

It computes a conditional probability from the "inverse" conditional probability. In a diagnosis setting:

$$\underbrace{P(\mathit{Problem} \mid \mathit{Symptoms})}_{\text{posterior}} = \frac{\overbrace{P(\mathit{Symptoms} \mid \mathit{Problem})}^{\text{likelihood}}\ \overbrace{P(\mathit{Problem})}^{\text{prior}}}{\underbrace{P(\mathit{Symptoms})}_{\text{evidence}}}$$

with

$$P(\mathit{Symptoms}) = \sum_{p \in \mathit{Val}(\mathit{Problem})} P(\mathit{Symptoms} \mid p)\, P(p).$$

Bayes' rule (after the Reverend Thomas Bayes, 1702–1761, whose essay was published posthumously in 1763) is one of the most fundamental tools in scientific data analysis and probabilistic reasoning. The reason is that knowledge usually comes in the "causal" direction — how often does flu produce muscle pain? — while questions come in the "diagnostic" direction: this person has muscle pain, so do they have flu? The names of the four parts:

- **Prior** $P(\mathit{Problem})$: belief in the problem *before* looking at the current case.
- **Posterior** $P(\mathit{Problem} \mid \mathit{Symptoms})$: belief *after* observing the symptoms.
- **Likelihood** $P(\mathit{Symptoms} \mid \mathit{Problem})$: how likely the problem is to produce these symptoms.
- **Evidence** $P(\mathit{Symptoms})$: how likely these symptoms are at all. By the law of total probability it is the sum above, so it can always be computed from priors and likelihoods.

Does the person with muscle pain have the flu?

Suppose we know (or believe) that $P(\mathit{mp} \mid \mathit{flu}) = 0.5$ (half of all flu patients have muscle pain), $P(\mathit{flu}) = 0.004$ (0.4 % of the population has the flu at any time) and $P(\mathit{mp}) = 0.1$ (10 % of people have muscle pain). Then

$$P(\mathit{flu} \mid \mathit{mp}) = \frac{P(\mathit{mp} \mid \mathit{flu})\, P(\mathit{flu})}{P(\mathit{mp})} = \frac{0.5 \times 0.004}{0.1} = 0.02.$$

Only 2 %! Counting people makes this intuitive. Out of 100 000 people, 400 have the flu, and 200 of them have muscle pain. But 10 000 people have muscle pain in total, so only $200/10\,000 = 2\,\%$ of those with muscle pain have the flu. The tiny prior dominates.

Do not flip the conditional

$P(\mathit{mp} \mid \mathit{flu}) = 0.5$ but $P(\mathit{flu} \mid \mathit{mp}) = 0.02$: the two directions can be wildly different, and confusing them is the classic mistake. Also note that a likelihood $P(\mathit{Symptoms} \mid \mathit{Problem})$ is a distribution over *symptoms*, not over problems: summing it over all problems does *not* have to give one.

2.5 Probabilistic Inference: Queries and Enumeration

Here is the scenario for the rest of the chapter. The full joint distribution over $\mathcal{X}$ is the system's *knowledge base*: its model of the world, assigning a probability to every possible state. The system's job is to compute, from this knowledge base and the observed values of some variables, the probability of any hypothesis we are interested in. Deriving the truth or probability of a hypothesis from a knowledge base and observations is called *inference* or *reasoning*. For us it almost always means computing (conditional) distributions over the variables of interest.

Probabilistic query

A *probabilistic query* asks for the conditional distribution

$$P(\mathbf{X} \mid \mathbf{E} = \mathbf{e}), \quad \text{or } P(\mathbf{X} \mid \mathbf{e}) \text{ for short,}$$

for a set of *query variables* $\mathbf{X} \subseteq \mathcal{X}$, a set of *evidence variables* $\mathbf{E} \subseteq \mathcal{X}$ and their observed values $\mathbf{e}$. In words: given that we have observed $\mathbf{e}$, how probable is each value combination of $\mathbf{X}$? All remaining variables are *hidden*.

With table (2.1) we could ask for an event probability, $P(\mathit{fever} \wedge \neg \mathit{mp})$ (someone has a fever but no muscle pain); a marginal distribution, $P(\mathit{Flu})$; or a conditional distribution, $P(\mathit{Flu} \mid \mathit{fever}, \neg \mathit{mp})$. All of them are answered the same way.

2.5.1 Inference by enumeration

Event probabilities are just sums of matching cells:

$$P(\neg \mathit{fever}) = \sum_{x,y} P(\neg \mathit{fever}, \mathit{MP} = x, \mathit{Flu} = y) = 0.10 + 0.03 + 0.04 + 0.50 = 0.67,$$
$$P(\mathit{fever} \wedge \neg \mathit{mp}) = \sum_{x \in \mathit{Val}(\mathit{Flu})} P(\mathit{fever}, \neg \mathit{mp}, \mathit{Flu} = x) = 0.10 + 0.05 = 0.15.$$

Conditional probabilities are a ratio of two such sums:

$$P(\mathit{flu} \mid \neg \mathit{fever}) = \frac{P(\mathit{flu} \wedge \neg \mathit{fever})}{P(\neg \mathit{fever})} = \frac{0.10 + 0.03}{0.10 + 0.03 + 0.04 + 0.50} = \frac{0.13}{0.67} \approx 0.194,$$
$$P(\neg \mathit{flu} \mid \neg \mathit{fever}) = \frac{P(\neg \mathit{flu} \wedge \neg \mathit{fever})}{P(\neg \mathit{fever})} = \frac{0.04 + 0.50}{0.67} = \frac{0.54}{0.67} \approx 0.806.$$

Look at the denominators: both are $P(\neg \mathit{fever}) = 0.67$, because the evidence probability does not depend on the value of the query variable. In general, when computing $P(X \mid y)$ we would compute the same $P(y)$ $|\mathit{Val}(X)|$ times. There is a smarter way: $P(y)$ is just the constant that makes the result sum to one, a *normalisation constant* Z, and we can get it for free afterwards by adding up the unnormalised numbers.

Inference by enumeration

To compute $P(\mathbf{X} \mid \mathbf{e})$, let $\mathbf{Z} = \mathcal{X} \setminus (\mathbf{X} \cup \mathbf{E})$ be the hidden variables.

1. **Sum out the hidden variables.** For every $\mathbf{x} \in \mathit{Val}(\mathbf{X})$,
$$P(\mathbf{x}, \mathbf{e}) = \sum_{\mathbf{z} \in \mathit{Val}(\mathbf{Z})} P(\mathbf{x}, \mathbf{e}, \mathbf{z}).$$
2. **Compute the normalisation constant** by adding up the results of step 1 (law of total probability):
$$Z = P(\mathbf{e}) = \sum_{\mathbf{x}} P(\mathbf{x}, \mathbf{e}).$$
3. **Renormalise:**
$$P(\mathbf{X} \mid \mathbf{e}) = \frac{1}{Z} P(\mathbf{X}, \mathbf{e}).$$

Two queries, step by step

Query $P(\mathit{Flu} \mid \neg \mathit{fever})$; the hidden variable is MP. Vectors list $(\mathit{flu}, \neg \mathit{flu})$ top to bottom:

$$P(\mathit{Flu}, \neg \mathit{fever}) = \sum_{x \in \{\mathit{mp}, \neg \mathit{mp}\}} P(\mathit{Flu}, \neg \mathit{fever}, x) = \begin{pmatrix} 0.10 + 0.03 \\ 0.04 + 0.50 \end{pmatrix} = \begin{pmatrix} 0.13 \\ 0.54 \end{pmatrix},$$

$$Z = 0.13 + 0.54 = 0.67, \qquad P(\mathit{Flu} \mid \neg \mathit{fever}) = \frac{1}{Z} \begin{pmatrix} 0.13 \\ 0.54 \end{pmatrix} \approx \begin{pmatrix} 0.194 \\ 0.806 \end{pmatrix}.$$

Query $P(\mathit{Flu} \mid \mathit{fever}, \neg \mathit{mp})$; nothing is hidden, so step 1 just reads off two cells:

$$P(\mathit{Flu}, \mathit{fever}, \neg \mathit{mp}) = \begin{pmatrix} 0.10 \\ 0.05 \end{pmatrix}, \quad Z = 0.15, \quad P(\mathit{Flu} \mid \mathit{fever}, \neg \mathit{mp}) \approx \begin{pmatrix} 0.667 \\ 0.333 \end{pmatrix}.$$

A fever without muscle pain makes flu twice as likely as no flu.

Always label the entries of a distribution

The slides write the same result once as $P(\mathit{Flu} \mid \neg \mathit{fever}) = \{0.806, 0.194\}$ (order $\neg \mathit{flu}, \mathit{flu}$) and once as $\{0.194, 0.806\}$ (order $\mathit{flu}, \neg \mathit{flu}$). Neither is wrong, but an unlabeled vector is ambiguous. On the exam, either label the entries or stick to the convention x^0 (false) first.

Inference by enumeration is completely general: any query, any evidence, one algorithm. Its problem is cost. With n variables of d values each, the joint table has d^n entries, and the sum in step 1 may have to touch almost all of them. Exponential space *and* exponential time — this is the first concrete appearance of the complexity problems from Section 1.4, and the reason the next sections are about independence.

2.5.2 Warm-up: everything you can ask of a small table

The slides close this part with a self-test (motivated by a tutorial by Ali Taylan Cemgil). It introduces a few quantities — maximum, mode and max-marginal — that will come back when we look for the *most likely* explanation of some evidence (Chapter 7).

Self-test with $X_1, X_2 \in \{1, 2\}$

$P(X_1, X_2)$	$X_2 = 1$	$X_2 = 2$
$X_1 = 1$	0.3	0.2
$X_1 = 2$	0.1	0.4

- **Marginals:** $P(X_1) = (0.5,\ 0.5)$ (row sums); $P(X_2) = (0.4,\ 0.6)$ (column sums).
- **Conditionals:** divide each column by its sum for $P(X_1 \mid X_2)$, each row by its sum for $P(X_2 \mid X_1)$:

$P(X_1 \mid X_2)$	$X_2 = 1$	$X_2 = 2$
$X_1 = 1$	0.75	0.33
$X_1 = 2$	0.25	0.67

$P(X_2 \mid X_1)$	$X_2 = 1$	$X_2 = 2$
$X_1 = 1$	0.6	0.4
$X_1 = 2$	0.2	0.8

- **Joint vs. posterior:** $P(X_1, X_2 = 2) = (0.2,\ 0.4)$ is a slice of the table and does *not* sum to one; $P(X_1 \mid X_2 = 2) = (0.33,\ 0.67)$ is that slice, normalised.
- **Evidence:** the normaliser of that slice, $P(X_2 = 2) = \sum_{x_1} P(x_1, X_2 = 2) = 0.6$.
- **Normalisation constant** of the whole table: $P(\Omega) = \sum_{x_1} \sum_{x_2} P(x_1, x_2) = 1$.
- **Max:** $\max_{x_1} P(x_1 \mid X_2 = 1) = 0.75$ — the *value* of the best probability.
- **Mode:** $x_1^* = \arg\max_{x_1} P(x_1 \mid X_2 = 1) = 1$ — *which* value is best.
- **Max-marginal:** $\max_{x_1} P(x_1, X_2) = (0.3,\ 0.4)$, the "skyline" of the table: for every x_2, the largest entry in its column. Unlike a marginal it maximises instead of summing.
- **Independent?** No: $P(X_1 = 1 \mid X_2 = 1) = 0.75 \neq 0.5 = P(X_1 = 1)$. Equivalently, $P(1, 1) = 0.3 \neq P(X_1 = 1)\, P(X_2 = 1) = 0.2$.

2.6 Independence

Normally, learning that one event happened changes our belief in another: $P(\alpha \mid \beta) \neq P(\alpha)$. Sometimes it does not. Toss two coins: whether the second one shows heads is not affected at all by what the first one showed. That is independence.

Independence of events

Two events α and β are *independent* in P, written $(\alpha \perp \beta)$, if (equivalently)

$$P(\alpha \mid \beta) = P(\alpha), \qquad P(\beta \mid \alpha) = P(\beta), \qquad P(\alpha \cap \beta) = P(\alpha)\, P(\beta).$$

Independence is symmetric: $(\alpha \perp \beta) \Leftrightarrow (\beta \perp \alpha)$.

The slides leave the equivalence as an exercise; it is two lines (assuming $P(\alpha), P(\beta) > 0$). If $P(\alpha \mid \beta) = P(\alpha)$, the product rule gives $P(\alpha \cap \beta) = P(\alpha \mid \beta)\, P(\beta) = P(\alpha)\, P(\beta)$. If $P(\alpha \cap \beta) = P(\alpha)\, P(\beta)$, then $P(\beta \mid \alpha) = P(\alpha \cap \beta)/P(\alpha) = P(\beta)$, and in the same way $P(\alpha \mid \beta) = P(\alpha)$. The product form is symmetric in α and β, which proves the symmetry too.

Independence of random variables (and sets of them)

Two sets of random variables $\mathbf{X}$ and $\mathbf{Y}$ are *independent*, written $(\mathbf{X} \perp \mathbf{Y})$, if for *all* value combinations $\mathbf{x} \in \mathit{Val}(\mathbf{X})$, $\mathbf{y} \in \mathit{Val}(\mathbf{Y})$:

$$P(\mathbf{x} \,|\, \mathbf{y}) = P(\mathbf{x}) \quad \text{and} \quad P(\mathbf{y} \,|\, \mathbf{x}) = P(\mathbf{y}),$$

in short $P(\mathbf{X} \,|\, \mathbf{Y}) = P(\mathbf{X})$ and $P(\mathbf{Y} \,|\, \mathbf{X}) = P(\mathbf{Y})$. Consequence: $P(\mathbf{X}, \mathbf{Y}) = P(\mathbf{X})\, P(\mathbf{Y})$. Single variables are the special case $\mathbf{X} = \{X\}$, $\mathbf{Y} = \{Y\}$.

Why does this matter so much? Because then every entry of the joint can be rebuilt from two smaller tables, $P(\mathbf{x}, \mathbf{y}) = P(\mathbf{x})\, P(\mathbf{y})$, so we only need to store distributions over $\mathbf{X}$ and over $\mathbf{Y}$ separately instead of over $\mathbf{X} \cup \mathbf{Y}$. Since table size grows exponentially with the number of variables, that is a big saving in space.

The extreme case: N independent coins $X_1, \ldots, X_N$. Then

$$P(X_1, \ldots, X_N) = P(X_1)\, P(X_2) \cdots P(X_N).$$

The joint has 2^N entries, but each $P(X_i)$ has only two (and really only one number, since $P(\text{tails}) = 1 - P(\text{heads})$). Storing $2N$ — or N — numbers instead of 2^N is an exponential reduction from $\mathcal{O}\left(2^N\right)$ to $\mathcal{O}(N)$.

Checking independence: one cell can only disprove it

Variable independence must hold for *every* combination of values. A single cell where $P(x, y) \neq P(x)\, P(y)$ proves dependence; a single cell where the product happens to match proves nothing. In table (2.1), for example, $P(\mathit{flu}, \mathit{mp}) = 0.25$ but $P(\mathit{flu})\, P(\mathit{mp}) = 0.38 \times 0.32 \approx 0.12$, so *Flu* and *MP* are dependent — as you would expect.

2.7 Conditional Independence

Full independence is rare in interesting worlds; things *are* connected. What is common is a weaker, and much more useful, property. Take

α = "patient Smith has a fever", β = "patient Smith has muscle pain", γ = "patient Smith has the flu".

Fever and muscle pain are dependent: they often go together, so learning about muscle pain raises our expectation of fever, $P(\alpha \,|\, \beta) > P(\alpha)$. But suppose we already *know* that Smith has the flu. Then learning about her muscle pain tells us nothing new about the fever — the fever is already "explained" by the flu: $P(\alpha \,|\, \beta \cap \gamma) = P(\alpha \,|\, \gamma)$.

Conditional independence

Two events α, β are *conditionally independent given* γ, written $(\alpha \perp \beta \,|\, \gamma)$, if (equivalently)

$$P(\alpha \,|\, \beta \cap \gamma) = P(\alpha \,|\, \gamma), \qquad P(\beta \,|\, \alpha \cap \gamma) = P(\beta \,|\, \gamma), \qquad P(\alpha \cap \beta \,|\, \gamma) = P(\alpha \,|\, \gamma)\, P(\beta \,|\, \gamma).$$

For sets of variables, $(\mathbf{X} \perp \mathbf{Y} \,|\, \mathbf{Z})$ if for all $\mathbf{x}, \mathbf{y}, \mathbf{z}$: $P(\mathbf{x} \,|\, \mathbf{y}, \mathbf{z}) = P(\mathbf{x} \,|\, \mathbf{z})$ and $P(\mathbf{y} \,|\, \mathbf{x}, \mathbf{z}) = P(\mathbf{y} \,|\, \mathbf{z})$, in short $P(\mathbf{X} \,|\, \mathbf{Y}, \mathbf{Z}) = P(\mathbf{X} \,|\, \mathbf{Z})$. Consequence: $P(\mathbf{X}, \mathbf{Y} \,|\, \mathbf{Z}) = P(\mathbf{X} \,|\, \mathbf{Z})\, P(\mathbf{Y} \,|\, \mathbf{Z})$. Conditional independence is symmetric as well.

The equivalence proof is the same as for plain independence, with "$\mid \gamma$" carried along everywhere. Now comes the step the entire course is built on:

Factorising with conditional independence

If $(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z})$, the joint distribution splits into smaller pieces:

$$\begin{aligned} P(\mathbf{X}, \mathbf{Y}, \mathbf{Z}) &= P(\mathbf{X} \mid \mathbf{Y}, \mathbf{Z})\, P(\mathbf{Y}, \mathbf{Z}) && \text{chain rule} \\ &= P(\mathbf{X} \mid \mathbf{Y}, \mathbf{Z})\, P(\mathbf{Y} \mid \mathbf{Z})\, P(\mathbf{Z}) && \text{chain rule} \\ &= P(\mathbf{X} \mid \mathbf{Z})\, P(\mathbf{Y} \mid \mathbf{Z})\, P(\mathbf{Z}) && \text{conditional independence.} \end{aligned}$$

Make sure you understand these three lines — they are the most fundamental idea in graphical models. The chain rule always works but leaves long conditioning sets; conditional independence lets us *delete* variables from them. A Bayesian network (Chapter 3) is nothing but this trick applied systematically to every variable.

Fever and muscle pain, with numbers

Build a joint over $\mathit{Flu}, \mathit{Fever}, \mathit{MP}$ from the factorisation $P(\mathit{Flu})\, P(\mathit{Fever} \mid \mathit{Flu})\, P(\mathit{MP} \mid \mathit{Flu})$ with

$$P(\mathit{flu}) = 0.1,$$
$$P(\mathit{fever} \mid \mathit{flu}) = 0.8, \quad P(\mathit{fever} \mid \neg \mathit{flu}) = 0.1, \qquad P(\mathit{mp} \mid \mathit{flu}) = 0.7, \quad P(\mathit{mp} \mid \neg \mathit{flu}) = 0.2.$$

Marginally dependent. By total probability $P(\mathit{fever}) = 0.1 \cdot 0.8 + 0.9 \cdot 0.1 = 0.17$, $P(\mathit{mp}) = 0.1 \cdot 0.7 + 0.9 \cdot 0.2 = 0.25$, and $P(\mathit{fever}, \mathit{mp}) = 0.1 \cdot 0.8 \cdot 0.7 + 0.9 \cdot 0.1 \cdot 0.2 = 0.074$. Hence

$$P(\mathit{fever} \mid \mathit{mp}) = 0.074/0.25 = 0.296 \; > \; 0.17 = P(\mathit{fever}).$$

Muscle pain is evidence for flu, and flu is evidence for fever.

Conditionally independent. Once flu is known, muscle pain adds nothing:

$$P(\mathit{fever} \mid \mathit{mp}, \mathit{flu}) = 0.8 = P(\mathit{fever} \mid \mathit{flu}).$$

Saving. The factorised form needs $1 + 2 + 2 = 5$ numbers instead of the $2^3 - 1 = 7$ of a general joint. Tiny here, enormous for big models.

Intuition & Mental Model

Once you know the cause, its effects stop "talking" to each other. All of the correlation between fever and muscle pain was routed *through* the flu; fixing the flu cuts that route. This is why causal stories are such a good guide to independencies: effects of a common cause are independent given that cause.

Independence and conditional independence do not imply each other

- **Dependent, but independent given Z:** fever and muscle pain above (common cause flu).
- **Independent, but dependent given Z:** toss two fair coins independently and let Z = "both coins show the same face". Given Z, the first coin *determines* the second, so they are dependent. We will meet this pattern as "explaining away" in Chapter 3.

2.8 Continuous Random Variables

2.8.1 Probabilities versus densities

So far every variable had a finite domain. Real applications, however, are full of continuous quantities: range sensors, temperature and pressure, GPS, camera pixels. And here a problem appears: each individual value has probability zero. Take $Val(X) = [0, 1]$ and give every number the same probability. There are infinitely many numbers, so if that probability were any positive value, the total would be infinite. Hence $P(X = x) = 0$ for all x. (The same happens for non-uniform distributions.) We need a different object.

Probability density and cumulative distribution

A function $p : \mathbb{R} \to \mathbb{R}$ is a *probability density function* (PDF) for a continuous variable X if $p(x) \geq 0$ for all $x \in Val(X)$, $p(x) = 0$ for all $x \notin Val(X)$, and $\int_{-\infty}^{+\infty} p(x)\,\mathrm{d}x = 1$. The *cumulative distribution function* (CDF) is

$$P(X \leq a) = \int_{-\infty}^{a} p(x)\,\mathrm{d}x,$$

so that

$$P(a \leq X \leq b) = P(X \leq b) - P(X \leq a) = \int_{a}^{b} p(x)\,\mathrm{d}x.$$

A PDF is *not* a probability distribution, and $p(x)$ is *not* a probability — it can even be larger than one. Probabilities live on intervals and are computed by integrating the density. The best mental picture: $p(x)$ is probability *per unit length*, the rate at which the CDF grows at x. For a small interval of width δ,

$$P(x \leq X \leq x + \delta) \approx p(x)\,\delta.$$

2.8.2 The uniform distribution

X is *uniformly distributed* on $[a, b]$, written $X \sim Unif[a, b]$, if

$$p(x) = \begin{cases} \dfrac{1}{b-a} & a \leq x \leq b, \\ 0 & \text{otherwise.} \end{cases}$$

Figure 2.2 shows the PDF and the CDF. Note what happens if the interval is short: on $[0, 0.5]$ the density is 2 everywhere. That is perfectly legal — only the *area* under a PDF has to be one.

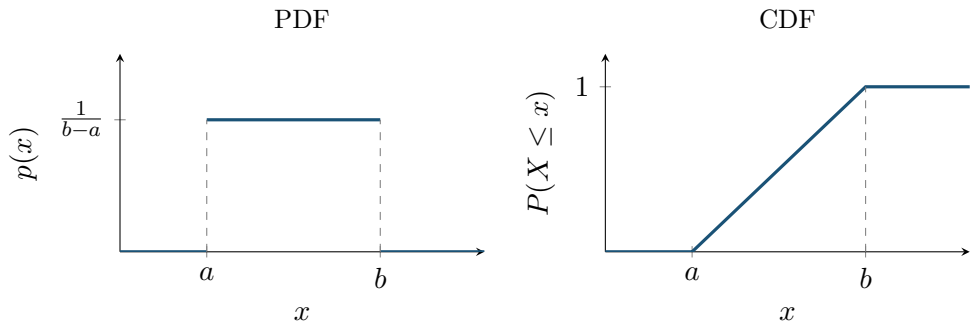


Figure 2.2: The uniform distribution on $[a, b]$. The CDF grows linearly from 0 to 1 across the interval; its slope is the density.

2.8.3 The normal (Gaussian) distribution

Normal distribution

X has a *normal (Gaussian) distribution* with mean μ and variance σ^2, written $X \sim \mathcal{N}(\mu, \sigma^2)$, if

$$p(x) = \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

The mean μ says where the bell is centred; the standard deviation σ says how wide it is (Fig. 2.3). A handy rule of thumb: about 68 % of the probability mass lies within $\mu \pm \sigma$, about 95 % within $\mu \pm 2\sigma$. Gaussians are everywhere in this course's temporal part, because sums of many small random disturbances tend to look Gaussian and because they are mathematically very convenient (Chapter 8).

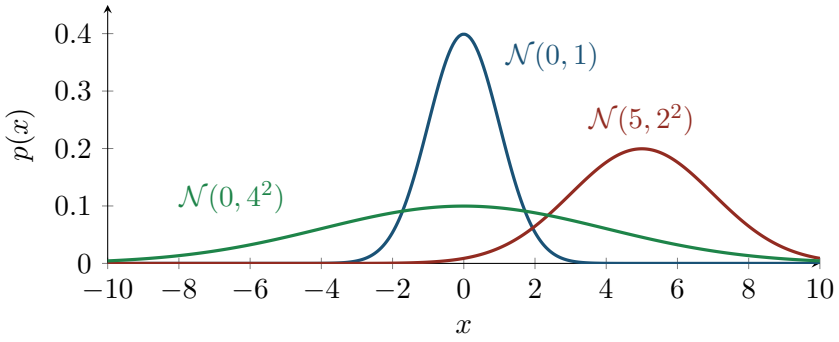


Figure 2.3: Three Gaussian densities. Wider bells are lower, because the area under each curve is always one.

2.8.4 The multivariate Gaussian

Multivariate normal distribution

N variables $\mathbf{X} = \{X_1, \ldots, X_N\}$ have a *joint (multivariate) normal distribution* with mean vector $\boldsymbol{\mu}$ and covariance matrix Σ, written $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, if

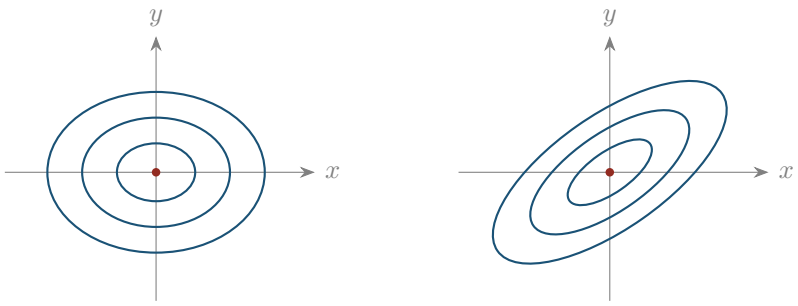
$$p(\mathbf{x}) = \frac{1}{(2\pi)^{N/2}\,|\Sigma|^{1/2}}\, \exp\Big[-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathsf{T}}\,\Sigma^{-1}\,(\mathbf{x}-\boldsymbol{\mu})\Big],$$

where $\boldsymbol{\mu} = (\mu_{X_1}, \ldots, \mu_{X_N})$, Σ is the $N \times N$ covariance matrix with $\Sigma_{ij} = \sigma_{ij}$ the covariance between X_i and X_j, and $|\Sigma|$ is its determinant.

In two dimensions the density is a "hill" whose contour lines are ellipses (Fig. 2.4). If X and Y are uncorrelated ($\sigma_{XY} = 0$), the ellipses are aligned with the axes. If they are positively correlated ($\sigma_{XY} > 0$), the ellipses tilt: large x tends to come with large y. The slides call σ_{ij} the "pairwise correlation"; strictly it is the *covariance* — the correlation coefficient is the covariance divided by $\sigma_i \sigma_j$ — but both are zero or non-zero together.

Uncorrelated vs. independent

For a *joint Gaussian*, uncorrelated means independent: if Σ is diagonal, the density factorises into a product of one-dimensional Gaussians. For general distributions this is false — zero correlation only rules out a *linear* relationship.



$\sigma_{XY} = 0$ $\sigma_{XY} > 0$

Figure 2.4: Contour lines of two 2-D Gaussians centred at $\boldsymbol{\mu} = \mathbf{0}$. Left: uncorrelated variables, axis-aligned ellipses. Right: positively correlated variables, tilted ellipses.

2.9 Chapter Checklist

After this chapter you should be able to:

- read the lecture's notation, and tell a probability from a distribution by looking at the argument of $P(\cdot)$;
- compute marginals, conditionals and event probabilities from a joint table, and use the chain rule, the law of total probability and Bayes' rule;
- state the probabilistic query $P(\mathbf{X} \mid \mathbf{e})$ and answer it by enumeration, including the normalisation trick;
- define (conditional) independence for events and for sets of variables, check it in a table, and explain why it makes representations smaller;
- explain the difference between a density and a probability, and work with uniform and (multivariate) Gaussian densities.

3

Bayesian Networks: Representation and Semantics

Chapter 2 ended with a promise and a problem. The promise: any query can be answered from the full joint distribution. The problem: the full joint is exponentially large. We also saw the way out — conditional independence lets us write the joint as a product of small pieces. A *Bayesian network* (BN) is the data structure that does this systematically: a directed acyclic graph that tells us *which* small pieces we need, plus one small table per variable that holds the numbers.

This chapter introduces BNs through one running example, the *student network*. We first play with it intuitively (which kinds of reasoning does it support?), then make its meaning precise in two equivalent ways — as a set of independence statements and as a factorised joint distribution — and finally look at how compact it is, how to build one, and how to handle continuous variables.

3.1 A Running Example: The Student Network

3.1.1 The story

The example comes from Koller & Friedman, with a small twist by the lecturer. A company wants to hire a college graduate. It is looking for genuinely interested and motivated people, but motivation cannot be tested directly. Instead, applicants hand in their SAT score (a standardised US university admission test) and their grade in a relevant university course. The company wants to use these two pieces of evidence to estimate whether an applicant is highly motivated. Our job: turn this into a quantitative model that algorithms can compute with.

Step 1: variables. Three concepts matter for the decision, each modelled as a discrete random variable with a deliberately coarse domain:

Interest/Motivation (I)	$Val(I) = \{lo, hi\}$	$= \{i^0, i^1\}$
SAT score (S)	$Val(S) = \{lo, hi\}$	$= \{s^0, s^1\}$
Grade (G)	$Val(G) = \{A, B, C\}$	$= \{g^1, g^2, g^3\}$

Careful with the grade: g^1 is the *best* grade (A) and g^3 the worst (C).

Step 2: dependency structure. The SAT score depends (among other things) on motivation, and so does the course grade. The story then grows a little. The grade also depends on how *difficult* the course was. And the student asks the professor for a recommendation *letter*, which the professor writes based on the course grade only — there are too many students to remember

anyone personally. Two more variables:

$$\begin{array}{lll} \text{Difficulty } (D) & Val(D) = \{easy, hard\} & = \{d^0, d^1\} \\ \text{Letter } (L) & Val(L) = \{weak, strong\} & = \{l^0, l^1\} \end{array}$$

Drawing an arrow from each cause to its effect gives the graph in Fig. 3.1: $D \to G \leftarrow I$, $I \to S$, $G \to L$.

Step 3: uncertainty. None of these relationships is deterministic. Motivation influences the SAT score but is not the only factor (and some SAT tests are easier than others). Motivation and difficulty do not uniquely determine the grade. An absent-minded professor might write a strong letter despite a poor grade. We model all of this with probabilities: every variable gets a *conditional probability distribution* (CPD) given its parents.

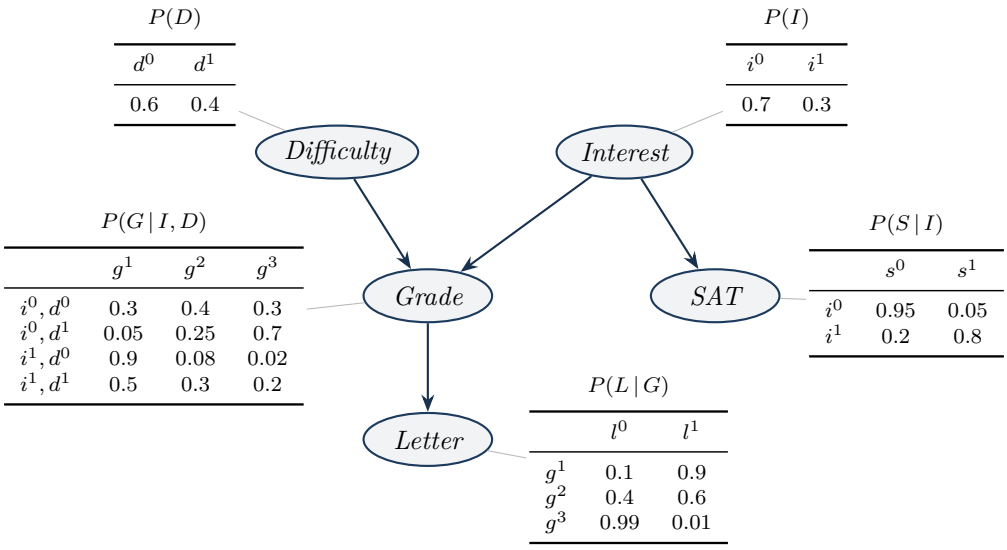


Figure 3.1: The student network with its conditional probability distributions (numbers from the lecture, after Koller & Friedman). Each row of each table is a probability distribution over the child, for one combination of parent values.

3.1.2 How to read a CPD

Variables without parents (D and I) get an ordinary, unconditional distribution. $P(I) = (0.7, 0.3)$ says that a random student is highly motivated with probability $P(i^1) = 0.3$. Variables with parents get one distribution *per combination of parent values*. In $P(S \mid I)$, the row for i^1 says that a motivated student gets a low SAT score with probability $P(s^0 \mid i^1) = 0.2$ and a high one with probability 0.8. $P(G \mid I, D)$ has four rows because its two parents have $2 \times 2 = 4$ value combinations.

Rows sum to one, columns do not

Every *row* of a CPD is a distribution over the child and must sum to one: $0.3+0.4+0.3 = 1$ for (i^0, d^0). A *column*, like the g^1 column $(0.3, 0.05, 0.9, 0.5)$, is a list of probabilities of the same child value under different conditions and has no reason to sum to anything in particular.

In the rest of the chapter we will see that this little picture is a lot more than a sketch. Its structure supports different patterns of reasoning, it captures all the independencies of the story, it is a compact representation of the full joint distribution over all five variables — precisely *because* of those independencies — and therefore it can answer any probabilistic query.

3.2 Reasoning Patterns

Remember the probabilistic query $P(\mathbf{X} \mid \mathbf{e})$ from Section 2.5: given the observed values $\mathbf{e}$ of the evidence variables, what is the distribution over the query variables? Let us ask the student network a sequence of such queries about a student called Joe and watch how the answers move. For now, simply take the numbers as given; they all follow from the CPDs in Fig. 3.1 (we compute two of them by hand in Section 3.4.1, and Chapter 4 shows how to compute such queries efficiently). Table 3.1 collects them.

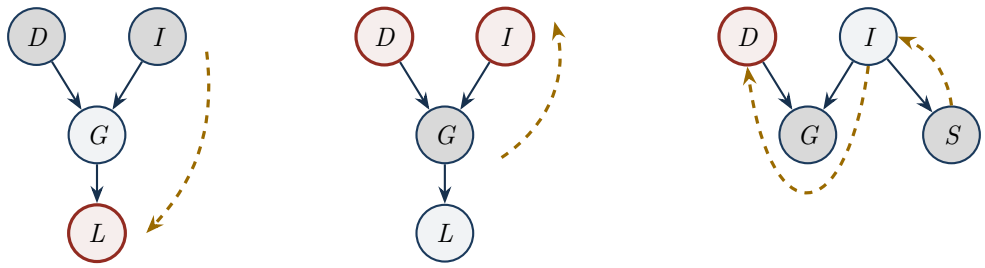
What we know about Joe	Query	Answer	Pattern
nothing	$P(l^1)$	0.502	—
not motivated	$P(l^1 \mid i^0)$	0.389	causal
not motivated, easy course	$P(l^1 \mid i^0, d^0)$	0.513	causal
nothing	$P(i^1)$	0.3	—
grade C	$P(i^1 \mid g^3)$	0.079	evidential
grade C	$P(d^1 \mid g^3)$	0.63	evidential
grade C, high SAT	$P(i^1 \mid g^3, s^1)$	0.578	intercausal
grade C, high SAT	$P(d^1 \mid g^3, s^1)$	0.76	intercausal (explaining away)

Table 3.1: Queries about Joe in the student network (rounded; all values re-computed from the CPDs). For comparison, the priors are $P(d^1) = 0.4$ and $P(i^1) = 0.3$.

Causal reasoning (prediction). Knowing nothing about Joe, his chance of a strong letter is about 0.502. When we learn that he is not very motivated (i^0), it drops to 0.389: unmotivated students tend to get worse grades, and worse grades mean weaker letters. When we *also* learn that the course was easy (d^0), it rises again to 0.513, because an easy course makes a good grade likely despite his low motivation. Reasoning from causes to effects — "downwards" in the network — is called *causal reasoning* or *prediction*.

Evidential reasoning (explanation). The prior probability that Joe is highly motivated is $P(i^1) = 0.3$, read directly from the model. Now we learn that he got a C (g^3). His probability of being motivated drops to 0.079. At the same time, the probability that the course was hard rises from 0.4 to 0.63: a C is also evidence for a hard course. Reasoning from observed effects to their possible causes — "upwards" — is called *evidential reasoning* or *explanation*.

Intercausal reasoning. Besides the C, Joe now hands in his SAT score, and surprisingly it is high (s^1). The probability that he is motivated jumps from 0.079 to 0.578. Why does the SAT outweigh the poor grade? Look at the CPDs: unmotivated students almost never get high SAT scores ($P(s^1 \mid i^0) = 0.05$), while motivated students can still get a C (especially in hard courses). But something else happens too: $P(d^1 \mid g^3, s^1) = 0.76$, up from 0.63. The SAT score tells us about motivation, and motivation — *together with the known grade* — tells us about difficulty. One cause of the grade (interest) gives us information about another cause of the grade (difficulty). A poor grade can be due to low motivation or to a hard course; once we learn that Joe is probably motivated, the hard course becomes the more plausible explanation. This is *intercausal reasoning*; its best-known special case is called *explaining away*.



(a) causal $P(l^1 \mid i^0, d^0)$ (b) evidential $P(i^1 \mid g^3)$, $P(d^1 \mid g^3)$ (c) intercausal $P(d^1 \mid g^3, s^1)$

Figure 3.2: The three reasoning patterns. Grey nodes are observed, red-framed nodes are queried, and the dashed arrows show the direction in which information flows. In (c), information about I reaches D only through the *observed* common effect G.

Intuition & Mental Model

The textbook picture of explaining away: your lawn is wet in the morning. Rain or the sprinkler? Both are plausible. Then you see that the sprinkler was on. The wet lawn is now *explained*, so rain becomes *less* likely, although rain and sprinkler have nothing to do with each other. The two causes were independent *until* we observed their common effect. The student example is the mirror image: evidence makes one explanation of the C (low motivation) *less* likely, so the other one (a hard course) becomes *more* likely. Same mechanism: two causes of an observed effect compete to explain it. This is the "independent, but dependent given Z" case from Section 2.7.

The remarkable thing is that we did not need a special algorithm for each pattern. All of them — including explaining away — fall out of one uniform inference procedure applied to the same model. A BN reasoner derives *every* consequence of *any* combination of observations. Nor do we need a notion of cause and effect for this to work: the arrows only describe how the model is *built*; information flows along them in both directions. In principle, dependencies can even be modelled in any direction, as long as the numbers are right — although causal models are usually much simpler (Section 3.6).

3.3 Semantics I: The Independencies Encoded by the Graph

We claimed that a BN is a model of the full joint distribution, and we know from Chapter 2 that independencies are what make a compact representation possible. So let us first figure out which independencies the student graph expresses. Recall the definition: $(\mathbf{X} \perp \mathbf{Y} \mid \mathbf{Z})$ means $P(\mathbf{x} \mid \mathbf{y}, \mathbf{z}) = P(\mathbf{x} \mid \mathbf{z})$ for all values, equivalently $P(\mathbf{X}, \mathbf{Y} \mid \mathbf{Z}) = P(\mathbf{X} \mid \mathbf{Z})\, P(\mathbf{Y} \mid \mathbf{Z})$ — the joint over $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$ factorises.

3.3.1 An intuitive analysis

Letter. According to the model, the letter depends *only* on the grade. If we know the grade and the professor's habits (the CPD of L), we can predict the letter, and learning anything else — motivation, SAT, difficulty — changes nothing: $(L \perp D, I, S \mid G)$. But L is *not* independent of the other variables if the grade is unknown: then motivation and difficulty do tell us something about the letter, via the grade (that was causal reasoning). So $(L \perp D, I, S \mid G)$ but $(L \not\perp D, I, S)$.

SAT. Motivation is the only direct influence on the SAT score. Once we know Joe's motivation, his grade, the course difficulty or the letter tell us nothing more about his SAT: $(S \perp D, G, L \mid I)$.

Without knowing I, however, a good grade or letter does make a high SAT more likely: $(S \not\perp G, L)$.

Grade. The grade depends directly only on interest and difficulty. Is it therefore independent of everything else given those two, $(G \perp L, S \mid I, D)$? **No!** Consider a strong student in a hard course (i^1, d^1). The chance of an A is medium: $P(g^1 \mid i^1, d^1) = 0.5$. If we now learn that the student got a strong letter (l^1), our belief in an A goes up: $P(g^1 \mid i^1, d^1, l^1) \approx 0.712$. The letter is an *effect* of the grade, so observing it is evidence about the grade, even when both parents are known: $(G \not\perp L \mid I, D)$.

What about G and S? If we know motivation and difficulty, we have certain expectations about the grade. Learning the SAT score could only change them by changing our belief about motivation — but motivation is already known. So $(G \perp S \mid I, D)$.

Difficulty and interest. These have no parents. Each of them can be influenced by information about its descendants (evidential or intercausal reasoning). But with no information about anything else, I tells us nothing about D, and vice versa; by the same argument S tells us nothing about D: $(D \perp I, S)$.

3.3.2 The general pattern: local independencies

All of these findings follow one rule. A variable's parents "shield" it from the influence of everything further up in the network. Given its parents, a variable is independent of all variables that are *not its descendants*. Information about its descendants, however, can still change our belief about it, through evidential reasoning.

Bayesian network structure (part I of the definition)

A *Bayesian network structure* $\mathcal{G}$ is a directed acyclic graph (DAG) whose nodes represent random variables $\mathcal{X} = \{X_1, \ldots, X_n\}$. The graph encodes the following *local independencies*:

$$(X_i \perp \mathrm{NonDesc}(X_i) \mid \mathrm{Pa}(X_i)) \qquad \text{for all } X_i \in \mathcal{X},$$

where $\mathrm{Pa}(X_i)$ are the parents of X_i in $\mathcal{G}$ and $\mathrm{NonDesc}(X_i)$ are the variables that are not descendants of X_i. In words: *every variable is conditionally independent of its non-descendants, given its parents.*

Figure 3.3 illustrates the definition for G, and Table 3.2 lists the local independencies of all five variables. They match the intuitive analysis exactly. All of them were also verified numerically against the joint distribution defined by the CPDs.

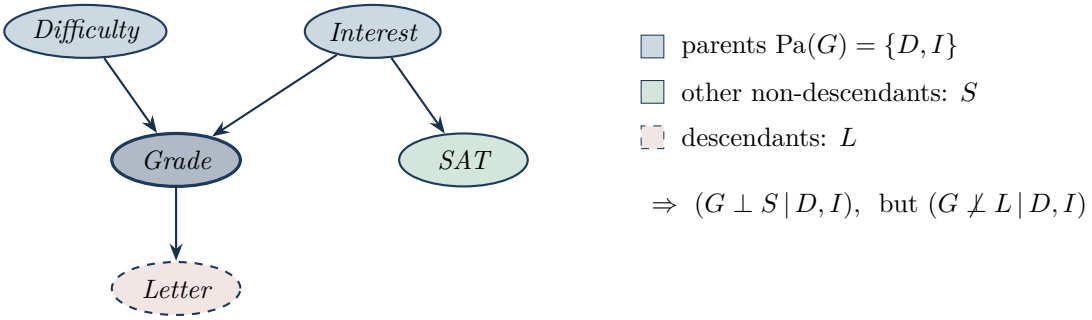


Figure 3.3: The local independence of G. Given its parents, G is independent of its other non-descendants, but not of its descendant L.

Variable	Pa	Descendants	Non-desc. (besides parents)	Local independence
D	—	G, L	I, S	$(D \perp I, S)$
I	—	G, S, L	D	$(I \perp D)$
G	D, I	L	S	$(G \perp S \mid D, I)$
S	I	—	D, G, L	$(S \perp D, G, L \mid I)$
L	G	—	D, I, S	$(L \perp D, I, S \mid G)$

Table 3.2: Local independencies of the student network. Conditioning on the parents is included in each statement; for a root node there is nothing to condition on.

Two things to watch out for

- **Descendants are excluded for a reason.** Observing a child (or grandchild) is evidence about its ancestors. This is exactly why $(G \not\perp L \mid I, D)$. A common mistake is to say "a node is independent of everything given its parents" — it is independent only of its *non-descendants*.
- **The list is not complete.** The local independencies are the ones the definition guarantees, but a graph usually implies *additional* ones that follow from them. Finding all of them requires a full independence analysis of the graph ("d-separation"), which the lecture lists under advanced topics.

3.4 Semantics II: A Factorised Joint Distribution

How do the CPDs relate to the overall joint distribution P over $\mathcal{X}$? The answer: if the graph correctly represents the conditional independencies of P, then P can be rebuilt as a product of the local CPDs. The argument is short and worth knowing.

Sketch of proof. Order the variables $X_1, \dots, X_n$ *topologically* with respect to the graph: every variable comes after all of its parents ("top to bottom"; for the student network, e.g. D, I, G, S, L). Such an ordering always exists in an acyclic graph. Now apply the chain rule in this order:

$$P(X_1, \dots, X_n) = P(X_1)\, P(X_2 \mid X_1)\, P(X_3 \mid X_1, X_2) \cdots = \prod_{i=1}^{n} P(X_i \mid X_1, \dots, X_{i-1}).$$

Because of the topological ordering, the predecessors $\{X_1, \dots, X_{i-1}\}$ contain all parents of X_i, possibly some other non-descendants, and *no* descendants. By the local independence $(X_i \perp \mathrm{NonDesc}(X_i) \mid \mathrm{Pa}(X_i))$, everything except the parents can be dropped from each factor:

$$P(X_i \mid X_1, \dots, X_{i-1}) = P(X_i \mid \mathrm{Pa}(X_i)).$$

Chain rule for Bayesian networks

$$P(X_1, \dots, X_n) = \prod_{i=1}^{n} P(X_i \mid X_1, \dots, X_{i-1}) = \prod_{i=1}^{n} P(X_i \mid \mathrm{Pa}(X_i)).$$

The CPDs jointly define a full probability distribution over $\mathcal{X}$; every entry of the joint is a product of one entry from each CPD.

Bayesian network (part II of the definition)

A *Bayesian network* $\mathcal{B}$ is a BN structure $\mathcal{G}$ in which every node X_i is associated with a CPD $P(X_i \mid \mathrm{Pa}(X_i))$. The network represents the joint distribution

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid \mathrm{Pa}(X_i)).$$

We say that P *factorises over* $\mathcal{G}$. A BN is a *factorised representation* of the joint distribution, and each CPD $P(X_i \mid \mathrm{Pa}(X_i))$ is called a *factor*.

Intuition & Mental Model

A BN can be read in two ways, and they describe the same object. Read as a graph, it is a list of independence statements ("given its parents, each variable ignores its non-descendants"). Read as a formula, it is a recipe for the joint distribution ("multiply one number from each table"). The proof above is the bridge: the independencies are exactly what allows the chain rule to collapse into the short product.

For the student network, the factorisation reads

$$P(D, I, G, S, L) = P(D)\, P(I)\, P(G \mid D, I)\, P(S \mid I)\, P(L \mid G).$$

One entry of the joint

What is the probability that a random student is not very motivated, takes an easy AI course, gets a B in it, has a low SAT score — and still gets a strong letter? Pick the matching entry from each CPD in Fig. 3.1:

$$\begin{aligned} P(d^0, i^0, g^2, s^0, l^1) &= P(d^0)\, P(i^0)\, P(g^2 \mid d^0, i^0)\, P(s^0 \mid i^0)\, P(l^1 \mid g^2) \\ &= 0.6 \times 0.7 \times 0.4 \times 0.95 \times 0.6 = 0.09576 \approx 9.6\,\%. \end{aligned}$$

3.4.1 Computing event probabilities from a BN

Since a BN *is* a full joint distribution, everything from Section 2.5 carries over: the probability of an event is the sum of all matching joint entries, and each joint entry is a product of CPD entries. Usually you do not even need to write out the whole joint — summing in a clever order is enough. Here are two of the numbers from Table 3.1.

The prior probability of a strong letter, $P(l^1)$

First get the marginal of G by summing over its parents (law of total probability, using $P(d, i) = P(d)\, P(i)$ since $D \perp I$):

$$\begin{aligned} P(g^1) &= \textstyle\sum_{i,d} P(i)\, P(d)\, P(g^1 \mid i, d) \\ &= 0.7{\cdot}0.6{\cdot}0.3 + 0.7{\cdot}0.4{\cdot}0.05 + 0.3{\cdot}0.6{\cdot}0.9 + 0.3{\cdot}0.4{\cdot}0.5 = 0.362, \\ P(g^2) &= 0.42{\cdot}0.4 + 0.28{\cdot}0.25 + 0.18{\cdot}0.08 + 0.12{\cdot}0.3 = 0.2884, \\ P(g^3) &= 1 - 0.362 - 0.2884 = 0.3496. \end{aligned}$$

Then sum over the grade:

$$P(l^1) = \sum_{g} P(g)\, P(l^1 \mid g) = 0.362 \cdot 0.9 + 0.2884 \cdot 0.6 + 0.3496 \cdot 0.01 \approx 0.502.$$

Evidential reasoning by hand, $P(i^1 \mid g^3)$

Bayes' rule with $P(g^3 \mid i) = \sum_d P(d)\, P(g^3 \mid i, d)$:

$$P(g^3 \mid i^1) = 0.6 \cdot 0.02 + 0.4 \cdot 0.2 = 0.092, \qquad P(g^3 \mid i^0) = 0.6 \cdot 0.3 + 0.4 \cdot 0.7 = 0.46,$$

$$P(i^1 \mid g^3) = \frac{P(g^3 \mid i^1)\, P(i^1)}{P(g^3 \mid i^1)\, P(i^1) + P(g^3 \mid i^0)\, P(i^0)} = \frac{0.092 \cdot 0.3}{0.092 \cdot 0.3 + 0.46 \cdot 0.7} = \frac{0.0276}{0.3496} \approx 0.079.$$

Note that the denominator is again $P(g^3) = 0.3496$.

Doing this by hand gets tedious quickly, and the order in which we sum things matters a lot for the effort. Organising these computations efficiently is the topic of Chapter 4.

3.5 Compactness

Recall the cost of the explicit joint. For N Boolean variables the full joint has 2^N entries, and specifying it takes $2^N - 1$ *independent parameters* — the last entry is redundant because everything must sum to one. Even for moderately sized worlds, a table like that is out of the question.

Now consider a *sparsely connected* BN over the same N Boolean variables, in which every variable has at most $k \ll N$ parents. Each row of a CPD needs one number: the probability of x_i^0; the probability of x_i^1 is one minus that. A variable with k parents has 2^k rows (one per parent configuration), so its CPD needs 2^k numbers, and all CPDs together need at most $N \cdot 2^k$ numbers. For fixed k that is *linear* in N: an exponential reduction from $\mathcal{O}\left(2^N\right)$ to $\mathcal{O}(N)$.

Counting independent parameters

For variables with arbitrary finite domains,

$$\#\text{params(full joint)} = \prod_{i=1}^{n} |\mathit{Val}(X_i)| - 1,$$

$$\#\text{params(BN)} = \sum_{i=1}^{n} \Big(|\mathit{Val}(X_i)| - 1 \Big) \prod_{Y \in \mathrm{Pa}(X_i)} |\mathit{Val}(Y)|\,.$$

Each CPD contributes (number of rows) $\times$ (number of child values -1). For Boolean variables with at most k parents: $\leq N \cdot 2^k$.

Counting parameters in the examples

Student network. There are $2 \cdot 2 \cdot 3 \cdot 2 \cdot 2 = 48$ atomic events, so the explicit joint needs 47 parameters. The BN needs

$$\underbrace{1}_{D} + \underbrace{1}_{I} + \underbrace{4 \cdot 2}_{G} + \underbrace{3 \cdot 1}_{L} + \underbrace{2 \cdot 1}_{S} = 15.$$

Ten independent coins. $2^{10} - 1 = 1023$ parameters for the explicit joint; as a BN with no edges, $10 \times 1 = 10$.

A realistic diagnosis model. A technical system with 50 Boolean component variables (ok/defective), each depending directly on at most 5 others: $2^{50} - 1 \approx 10^{15}$ parameters for the joint, but at most $50 \times 2^5 = 1600$ for the BN.

The diagnosis world of Fig. 1.1. $64 - 1 = 63$ for the joint. As a BN: *Season* has 4 values (3 parameters), *Flu* and *Hayfever* each have 4 rows (4 each), *MusclePain* has 2 rows (2), *Congestion* has 4 rows (4): $3 + 4 + 4 + 2 + 4 = 17$.

Subtract one per row, not one per table

In a CPD, *every row* is its own distribution, so every row has one redundant entry. A CPD for a three-valued child with two Boolean parents has $4 \times 3 = 12$ entries but $4 \times 2 = 8$ independent parameters — not 11. For the full joint, which is a single distribution, you subtract one in total.

3.6 Building a Bayesian Network

Constructing a BN means doing three things:

1. **Choose the variables and their domains.** This is always the system designer's job.
2. **Specify the structure.** Either by hand, with the designer and a domain expert working together (*knowledge engineering*), or automatically by learning it from example situations (*machine learning*, Chapter 5).
3. **Specify the CPDs.** Usually learned from data; in realistic models there are far too many numbers to elicit by hand.

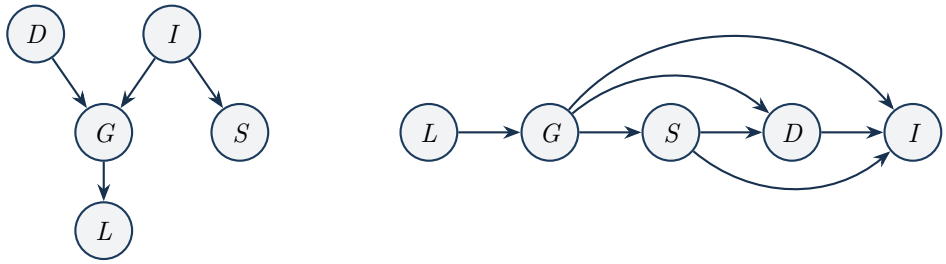
One piece of advice matters a lot: **model in the direction of causality**. Causal models are usually simpler, sparser and more intuitive than models built "backwards". The slides leave it as an exercise to check this for the student network; here is the solution.

The student network, built backwards

Suppose we insist on the anti-causal ordering L, G, S, D, I and, for each variable, choose as parents the smallest set of its predecessors that makes it independent of the remaining predecessors (so that the local independencies hold). Using the true distribution of the student network:

- L: no predecessors, no parents.
- G: depends on L, so $\mathrm{Pa}(G) = \{L\}$.
- S: given G it is independent of L, so $\mathrm{Pa}(S) = \{G\}$.
- D: independent of L given G, S — but *not* independent of S given G (explaining away: given the grade, S tells us about I, which tells us about D). So $\mathrm{Pa}(D) = \{G, S\}$.
- I: independent of L given the rest, but depends on G, S and D: $\mathrm{Pa}(I) = \{G, S, D\}$.

The result (Fig. 3.4) has 7 edges instead of 4 and needs $1 + 2 \cdot 2 + 3 \cdot 1 + 6 \cdot 1 + 12 \cdot 1 = 26$ parameters instead of 15. It represents the same distribution, but it is bigger and much harder to interpret — try to explain to a domain expert what $P(I \mid G, S, D)$ means! (All independence claims were checked numerically.)



causal order
4 edges, 15 parameters

anti-causal order L, G, S, D, I
7 edges, 26 parameters

Figure 3.4: The same distribution with two different structures. Building the network against the direction of causality forces extra edges.

3.6.1 Two larger examples

Figure 3.5 shows a simplified BN for car diagnosis (originally from lecture slides by Stuart Russell). The colours show the different roles that variables play in a diagnostic model: the *problem* to be explained (the car won't start), *tests* whose values can be obtained (lights, gauges, dipstick), possible *diagnoses* that are not directly observable (dead battery, no oil, broken starter, ...), and *auxiliary* variables. The auxiliary nodes, such as "battery flat" or "no charging", do not correspond to anything we would test or repair. They exist to make the model more structured and compact: many causes act on the tests *through* a flat battery, so they only need one shared intermediate node instead of separate edges to every test.

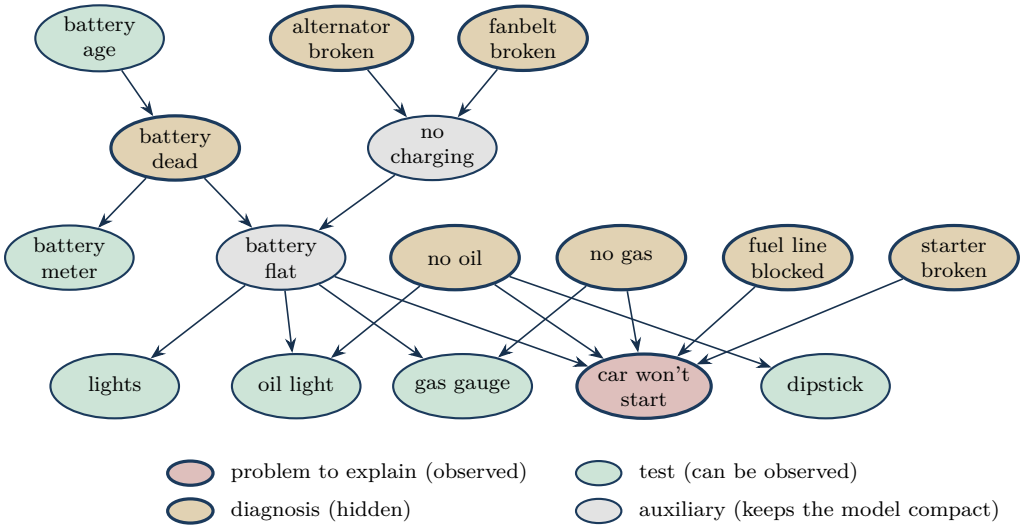


Figure 3.5: A simplified Bayesian network for car diagnosis (after S. Russell). Arrows point from causes to effects; "car won't start" is the problem to be explained.

The second example is a real one: the ALARM network for monitoring patients in an intensive care unit (Beinlich et al., 1989), with 37 variables. Despite its size, every variable depends directly on only a few others, which is exactly what keeps the model small enough to specify and to compute with.

Figure to insert: The ALARM network for ICU monitoring

A Bayesian network with 37 variables for monitoring ventilated patients (e.g. HYPOVOLEMIA, LVFAILURE, SAO2, HR, BP, VENTLUNG). Any clean rendering of the full graph works.

Search hint: "ALARM Bayesian network Beinlich 1989" — e.g. the graph on the bnlearn repository page for the ALARM network.

Figure 3.6: The ALARM network (Beinlich et al., 1989): a realistic medical Bayesian network with 37 variables, each with only a few parents.

3.7 Continuous Variables

So far all variables were discrete, which let us write every CPD $P(X \mid \mathrm{Pa}(X))$ as a table. But many systems involve continuous quantities — the position and velocity of an aircraft or a robot, and our measurements of them — which are best modelled by real-valued variables. There are two ways to deal with them.

Solution 1: discretisation. Split the domain of the variable into a fixed number of intervals and represent each value by the label of its interval. The advantage is that all of our machinery applies unchanged. The disadvantages are the information lost in the approximation and, above all, complexity. For reasonably accurate robot tracking we might need 1000 intervals per position coordinate, i.e. $1000 \times 1000 = 10^6$ different (x, y) positions. Now imagine a CPD with that as a parent.

Solution 2: continuous variables with continuous CPDs. Allow $Val(X) \subseteq \mathbb{R}$ and model CPDs as density functions $p(X \mid \mathrm{Pa}(X))$. This raises two questions: how do we model the distribution of a continuous variable, and how do we define $p(X \mid \mathrm{Pa}(X))$ when the parents are continuous, too? Take $X \to Y$ with both continuous. The CPD $p(Y \mid X)$ must contain one distribution $p(Y \mid x)$ for *every* value x — infinitely many. We cannot list them.

The trick is to pick a *parametric* family of densities and to make its parameters a *function* of the parent values. Then a few numbers describe all the infinitely many conditional distributions at once. The most important special case:

Linear Gaussian model

Let Y be a continuous variable with continuous parents $X_1, \ldots, X_k$. Y has a *linear Gaussian model* if there are parameters $\beta_0, \beta_1, \ldots, \beta_k$ and σ^2 such that

$$p(Y \mid x_1, \ldots, x_k) = \mathcal{N}(\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k;\ \sigma^2),$$

or in vector notation $p(Y \mid \mathbf{x}) = \mathcal{N}(\beta_0 + \boldsymbol{\beta}^{\mathsf{T}}\mathbf{x};\ \sigma^2)$. A *linear Gaussian Bayesian network* is a BN in which all variables are continuous and all of them have linear Gaussian models.

So every distribution over a single variable is a Gaussian; the *mean* of the child is a linear function of the parents; and the *variance* is fixed — it does not depend on the parents. For example, $p(Y \mid x) = \mathcal{N}(3.5x - 0.9;\ 1.0)$: whatever x is, Y is spread around $3.5x - 0.9$ with variance one. An equivalent and often more intuitive way to say this: Y is a linear function of its parents plus

Gaussian noise,

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2).$$

Figure 3.7 shows what this looks like for one parent.

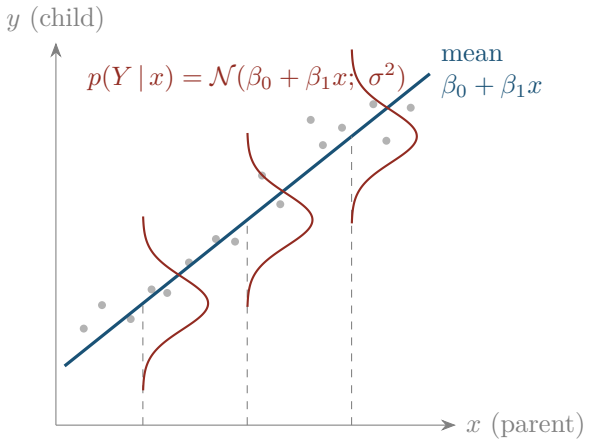


Figure 3.7: A linear Gaussian CPD with one parent. For every parent value x the child is Gaussian (red curves); the mean moves along a straight line, while the width stays the same everywhere. Grey dots are samples from the model.

A bicycle on a straight road

A cyclist rides along a straight road; once per second we model her position $X^{(t)}$ (in m) and her velocity $V^{(t)}$ (in m/s). Ideally, $X^{(t+1)} = X^{(t)} + V^{(t)}$: at metre 510.4 with a speed of 5 m/s, she should be at metre 515.4 one second later. But real motion is never perfectly precise, so it is more realistic to say that the next position is normally distributed around that value:

$$p(X^{(t+1)} \mid x^{(t)}, v^{(t)}) = \mathcal{N}(x^{(t)} + v^{(t)};\ 0.8).$$

This is a linear Gaussian model with $\beta_0 = 0$, $\beta_1 = \beta_2 = 1$ and $\sigma^2 = 0.8$. (The slides say "variance 0.8 meters"; strictly a variance is measured in m^2 — the standard deviation is $\sqrt{0.8} \approx 0.89$ m.)

Gaussian distributions, linear dependencies and a fixed variance are fairly severe restrictions. Some processes satisfy them and some do not. But they are reasonably good models of many physical phenomena, and — the real selling point — they support simple and elegant reasoning algorithms. We will meet a concrete one for a restricted family of such models in the chapter on the Kalman filter (Chapter 8).

Beyond this chapter. The lecture explicitly leaves out a full analysis of independence in BN graphs, algorithms that construct a graph for a given set of independencies, and other kinds of local models (context-specific CPDs, tree and rule CPDs, hybrid discrete–continuous models, logistic models, . . .).

3.8 Chapter Checklist

What you should take away from this chapter (following the lecture's own list):

- the probabilistic query, and the typical reasoning patterns: causal, evidential and intercausal (explaining away);

- the definition of a Bayesian network, in both parts: the graph with its local independencies, and the CPDs with the factorised distribution;
- which independencies a BN encodes — $(X_i \perp \mathrm{NonDesc}(X_i) \mid \mathrm{Pa}(X_i))$ — and why descendants are excluded;
- the chain rule for Bayesian networks and its proof via a topological ordering;
- how to compute event probabilities from a BN;
- the notion of independent (non-redundant) parameters, and how a BN makes the representation of the full joint compact;
- the basic idea of linear Gaussian models (details follow with the Kalman filter).

4

Inference in Bayesian Networks

Chapter 3 showed that a Bayesian network *is* a full joint distribution, so every query can in principle be answered by enumeration — multiply CPD entries, sum over the hidden variables, normalise. In practice that is exponential in the number of variables. This chapter is about doing it smarter: first exactly, by organising the sums cleverly, and then approximately, by *sampling* from the network when exact inference is too expensive.

Planned content: Lecture units VO 4 ("Bayes Nets: Exact & Approximate Inference (1)") and VO 5 ("Bayes Nets: Approximate Inference and MCMC"). The section list below is the expected structure; adapt it to the slides.

4.1 Exact Inference by Enumeration in a Bayesian Network

Planned content: Recap of the probabilistic query; enumeration with the factorised joint; pushing sums inwards; the student network as a worked example (re-using the numbers of Table 3.1); cost of naive enumeration.

4.2 Variable Elimination

Planned content: Factors and factor operations (product, summing out, reduction by evidence); the variable-elimination algorithm; worked example; the effect of the elimination order on cost; intermediate factor sizes. Figure (TikZ, to be drawn): a small elimination example showing which factors are created.

4.3 The Complexity of Exact Inference

Planned content: Exact inference is NP-hard in general; polytrees (singly connected networks) allow inference in linear time; why loops make it hard; motivation for approximate methods.

4.4 Approximate Inference by Stochastic Sampling

Planned content: The idea: estimate probabilities from frequencies in samples. Prior (forward) sampling in topological order; rejection sampling for conditional queries and its weakness with unlikely evidence; likelihood weighting; consistency and convergence of the estimates.

4.5 Markov Chain Monte Carlo and Gibbs Sampling

Planned content: Markov chains over complete states; Gibbs sampling: resample one variable at a time given its Markov blanket; stationary distribution; burn-in and mixing; comparison with likelihood weighting. Figure (TikZ, to be drawn): a chain of complete assignments where each step resamples one variable.

4.6 Chapter Checklist

Planned content: To be written together with the chapter.

5

Learning Bayesian Networks

So far somebody handed us the graph and all the CPDs. In reality, the numbers — and often the structure — have to come from data. This chapter addresses the third of the complexity problems from Chapter 1, modelling complexity: given a set of observations, find the parameters (and maybe the structure) of a Bayesian network that explains them well. Two schools of parameter estimation appear — maximum likelihood and Bayesian — followed by the question of how to avoid overfitting and how to search over structures.

Planned content: Lecture units VO 6 ("Learning (1): General Issues & ML Parameter Learning"), VO 7 ("Learning (2): Bayesian Parameter Learning") and VO 8 ("Learning (3): Overfitting, Regularisation, Structure Learning"). Expected structure below; adapt it to the slides.

5.1 The Learning Problem

Planned content: What is given and what is learned (parameters vs. structure); complete vs. incomplete data; the i.i.d. assumption; goals of learning (density estimation, prediction, knowledge discovery).

5.2 Maximum-Likelihood Parameter Learning

Planned content: The likelihood function; ML estimation for a single Bernoulli/ multinomial variable ("counting"); decomposition of the BN likelihood into independent per-CPD problems; sufficient statistics; worked example with a small data table; the zero-count problem.

5.3 Bayesian Parameter Learning

Planned content: Parameters as random variables; prior, posterior and predictive distribution; Beta and Dirichlet priors; conjugacy; pseudo-counts and their intuition; MAP vs. full Bayesian prediction; worked coin example. Figure (pgfplots, to be drawn): Beta priors and how the posterior sharpens with data.

5.4 Overfitting and Regularisation

Planned content: Why maximum likelihood overfits, especially for CPDs with many parents and little data; priors as regularisers; the bias–variance view; model complexity.

5.5 Structure Learning

Planned content: Score-based structure learning: likelihood score (and why it always prefers more edges), penalised scores (BIC/MDL), Bayesian score; local search over graphs (add/delete/reverse an edge); decomposability of scores.

5.6 Chapter Checklist

Planned content: To be written together with the chapter.

6

Temporal Models: Dynamic Bayesian Networks

Many of the examples from Chapter 1 — tracking aircraft, driving a car, following music — are about a world that *changes over time*. We observe it step by step and want to keep our beliefs up to date. Dynamic Bayesian networks extend the BN idea to such processes: the same small "template" of variables is repeated for every time step, with edges inside a time slice and from one slice to the next.

Planned content: Lecture unit VO 9 ("Dynamic Bayes Nets and State-Observation Models"). Expected structure below; adapt it to the slides.

6.1 Template Variables and Time Slices

Planned content: Discrete time, template variables $X_i^{(t)}$ (see Table 2.1); the Markov assumption; stationarity (time-invariant transition model); the two-slice representation (2-TBN) and unrolling it over T steps.

6.2 State-Observation Models

Planned content: Hidden state and observations; transition model $P(\mathbf{X}^{(t+1)} \mid \mathbf{X}^{(t)})$ and sensor/observation model $P(\mathbf{Y}^{(t)} \mid \mathbf{X}^{(t)})$; examples (robot localisation, the bicycle of Section 3.7). Figure (TikZ, to be drawn): an unrolled state-observation chain.

6.3 Inference Tasks over Time

Planned content: Filtering (belief state), prediction, smoothing, most likely explanation; the recursive forward update (predict, then condition on the new observation); why naive unrolling becomes expensive.

6.4 Chapter Checklist

Planned content: To be written together with the chapter.

7

Hidden Markov Models

A hidden Markov model is the simplest interesting state-observation model: a single discrete hidden state that evolves as a Markov chain, and an observation emitted at every step. Despite (or because of) this simplicity, HMMs power speech recognition, gene finding and — close to the lecturer's research — following a live music performance in a score. This chapter covers the classic inference algorithms, learning the parameters from data, and applications.

Planned content: Lecture units VO 10 ("Hidden Markov Models (1): Inference") and VO 11 ("Hidden Markov Models (2): Learning and Applications"). Expected structure below; adapt it to the slides.

7.1 The HMM as a Dynamic Bayesian Network

Planned content: States, observations, initial distribution π, transition matrix A, emission probabilities B; the HMM as a special DBN; a small running example. Figure (TikZ, to be drawn): HMM unrolled over time, plus the state transition diagram.

7.2 Filtering and Likelihood: The Forward Algorithm

Planned content: Forward messages α_t; recursion and normalisation; computing the likelihood of an observation sequence; complexity $\mathcal{O}(TN^2)$ instead of $\mathcal{O}(N^T)$.

7.3 Smoothing: The Forward–Backward Algorithm

Planned content: Backward messages β_t; combining forward and backward messages into posteriors over past states.

7.4 The Most Likely State Sequence: Viterbi

Planned content: Max instead of sum (compare the max-marginal of Section 2.5); dynamic programming over the trellis; backtracking; worked example. Figure (TikZ, to be drawn): Viterbi trellis with the best path highlighted.

7.5 Learning HMMs: Baum–Welch

Planned content: Supervised (counting) vs. unsupervised learning; the EM idea; expected counts from forward–backward; re-estimation formulas; local optima.

7.6 Applications

Planned content: Speech recognition; real-time music tracking / score following; other applications mentioned in the lecture.

Figure to insert: Real-time score following

An image of a score follower in action: a live audio signal aligned to the sheet music, with the current position highlighted (as in the lecture's video of real-time music following).

Search hint: "automatic score following live performance" or "real-time music tracking score alignment"; screenshots of the JKU Computational Perception group's demos fit well.

Figure 7.1: An HMM-based score follower tracks a live performance in the printed score.

7.7 Chapter Checklist

Planned content: To be written together with the chapter.

8

Continuous Linear Systems: The Kalman Filter

What an HMM does for a discrete hidden state, the Kalman filter does for a continuous one — provided the world is *linear* and *Gaussian*. This is exactly the linear Gaussian setting previewed in Section 3.7, and it is what makes the bicycle example tractable: if the belief about the current state is Gaussian and all dependencies are linear Gaussian, then the belief stays Gaussian forever, and filtering reduces to updating a mean and a covariance.

Planned content: Video lecture "Modelling Continuous Linear Systems: The Kalman Filter". Expected structure below; adapt it to the video.

8.1 Linear Gaussian State-Space Models

Planned content: Continuous state and observation variables; linear Gaussian transition and sensor models (connection to Section 3.7); the bicycle/robot tracking example.

8.2 The Predict–Update Cycle

Planned content: Prediction step (the belief spreads out); update step (conditioning on the observation sharpens it); the Kalman gain as a weighting between prediction and measurement; the 1-D case in detail, then the matrix form. Figure (pgfplots, to be drawn): prior, prediction, measurement and posterior Gaussians in 1-D.

8.3 Properties and Limitations

Planned content: Why Gaussians stay Gaussian; optimality under the model assumptions; what goes wrong for non-linear or multi-modal problems; extensions if covered in the video.

8.4 Chapter Checklist

Planned content: To be written together with the chapter.